

Disaggregating ESG

Heterogeneous Determinants of International Tourism Demand in Europe, 2000-2023

[Author 1]

[Author 2]

[Author 3]

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Introduction

Environmental, social, and governance considerations have moved from corporate finance into mainstream tourism policy. Destination management organisations, national tourism boards, and supranational agencies increasingly treat ESG progress as both a determinant and an outcome of tourism flows. Yet empirical evidence on whether and how ESG factors are associated with international tourist demand remains thin, fragmented, and methodologically uneven. The literature offers a mixture of cross-country regressions, case studies, and composite-score approaches, and conclusions vary substantially depending on which ESG dimensions are emphasised, how they are aggregated, and which countries are included.

i Literature gap paragraph (co-authors to populate)

Three points to establish. First, prior work tends to use either composite ESG scores that mask within-pillar heterogeneity or single-indicator analyses that miss the joint structure of ESG. Second, European panel evidence is limited and rarely tested for fragility across specification choices. Third, the heterogeneity between transition and frontier European economies is undertheorised in the tourism context.

This paper studies how disaggregated ESG factors are associated with international tourist arrivals in a panel of 27 European countries observed annually over 2000 to 2023. We address

three questions. First, which environmental, social, and governance factors are robustly associated with international arrivals once country and year fixed effects absorb time-invariant country traits and common annual shocks? Second, are these associations heterogeneous between transition and frontier European economies, and through which dimensions specifically? Third, how sensitive are these conclusions to specification choices, identification framework (within-country deviations versus long-run cointegrating relationships), and the treatment of cross-sectional dependence?

Our contribution is twofold. Substantively, we document (i) a long-run co-movement between governance reform (corruption control) and international arrivals identified off persistent country reform trajectories — a relationship that does *not* survive the inclusion of country-specific linear time trends and is therefore best read as a slow-moving institutional-trajectory result rather than a within-country structural deviation result; (ii) within-country associations between air quality (PM2.5 exposure), digital infrastructure (internet penetration), unemployment, and international arrivals; and (iii) substitution-style cross-sectional dependence in residuals after year fixed effects, suggesting that improvements in any one destination partly draw demand from competing European destinations. Methodologically, we show that the within-country versus long-run identification distinction matters quantitatively for tourism-ESG analysis, that conclusions depend on how the within-pillar redundancy in governance indicators is handled, and that pooled cross-sectional learners (PCA composites, elastic net, random forest) and within-country causal estimators answer different questions and accordingly point to different ESG dimensions.

The within-pillar redundancy we document for the WGI governance indicators in European panel data may apply more broadly to ESG measurement systems with similar internal structure. We do not claim, and the present analysis cannot establish, that this generalises beyond our setting; future work could test the redundancy structure in other ESG measurement contexts.

The paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 develops the theoretical framework and hypotheses. Section 4 describes the panel data and the cleaning pipeline. Section 5 sets out the empirical strategy and the diagnostic battery. Section 6 presents the results, including the new long-run, cross-sectional-dependence-robust, and imputation-uncertainty-propagating estimators. Section 7 discusses interpretation, mechanisms, and limitations. Section 8 concludes.

Literature Review

i Co-authors to populate

Each subsection should close with one paragraph stating explicitly the gap our paper addresses, so that the contribution claim in Section 1 maps directly onto identified gaps.

Governance, institutional quality, and international tourism

i Note

Cover prior empirical work linking corruption, rule of law, political stability, and government effectiveness to tourism flows. Note the heavy reliance on cross-country regressions, the relative shortage of within-country panel evidence, and within that literature the convention of using **control of corruption** rather than rule of law as the headline governance proxy in tourism applications. Cite specific tourism papers using corruption (Poprawe 2015; Saha and Yap 2014; Yap and Saha 2013; Das and DiRienzo 2010) so that the choice in our Section 4.5 has a literature anchor.

Environmental conditions and destination choice

i Note

Cover air quality, climate, biodiversity, and pollution as correlates of destination demand. Note the emerging literature on PM2.5 and tourism, the simultaneity / reverse-causality issue (tourist activity also generates emissions), and the gap regarding within-country identification.

Digital infrastructure and tourism mobility

i Note

Cover the role of internet penetration, digital booking platforms, and information frictions in international travel. Note the saturation hypothesis for frontier economies and the granger-causality evidence (one-way precedence) that supports a stronger interpretation here than for the other regressors.

ESG measurement, composite indices versus disaggregated indicators

i Note

Cover the methodological literature on ESG aggregation, the disagreement among rating providers, and the critique of composite scores. Position our theory-sparse approach against this background, and note that we do *not* claim our theory-sparse versus kitchen-sink contrast generalises beyond the WGI governance pillar in European panel data.

European tourism in panel: prior evidence and methodological precedents

i Note

Cover prior panel studies of European tourism demand, with attention to specifications, identification strategies, treatment of cross-sectional dependence, and findings.

Theoretical Framework and Hypotheses

International tourism demand responds to a constellation of destination characteristics that potential travellers either observe directly or infer from country-level signals. We organise these characteristics through the ESG taxonomy because that taxonomy increasingly structures the policy and information environment in which destination choices are made.

Three transmission channels link ESG factors to international arrivals. First, *reputation*. Visible governance failures, environmental degradation, or social distress shape destination perception in source markets. Second, *supply quality*. Better governance, cleaner environments, and stronger digital infrastructure raise the underlying quality of the tourism product. Third, *institutional friction*. Stable governance and developed digital infrastructure lower the transaction costs of international travel.

Five hypotheses follow. They differ in their identifying variation, and the differentiation matters quantitatively for what the data can support.

H1 (long-run trend co-movement). Governance quality, measured through corruption control, is positively associated with international arrivals along persistent within-country reform trajectories over 2000-2023. *Mechanism*. Sustained improvements in corruption control build destination reputation, lower friction in tourism services, and signal credible institutional reform to source markets and travel intermediaries. *Identifying variation*. H1 is identified off the long-run trajectory of corruption control within country, not off year-on-year deviations from that trajectory. The country-specific time-trends robustness in Section 6.7 (Table 18) confirms this: once persistent country reform trajectories are absorbed, residual year-on-year

deviations in corruption control display the *opposite* sign. H1 should therefore be read as a slow-moving institutional-trajectory result, not as a contemporaneous within-country deviation result.

H2. Air quality, captured through inverse PM2.5 exposure, is positively associated with international arrivals within country. *Mechanism.* Lower PM2.5 raises perceived destination quality and reduces health concerns. *Identifying variation.* The bidirectional Granger evidence (Appendix B) indicates a simultaneous-feedback structure; the contemporaneous coefficient mixes the structural channel with the reverse channel where tourist activity itself influences pollution. We interpret it as a co-movement, not a structural elasticity.

H3. Digital infrastructure, measured through internet penetration, is positively associated with international arrivals within country. *Mechanism.* Higher internet penetration lowers planning, booking, and on-site coordination costs. *Identifying variation.* The clean unidirectional Granger precedence (Appendix B) supports a within-country conditional-association reading with stronger temporal grounding than the other regressors.

H4. Labour market conditions, captured through inverse unemployment, are positively associated with international arrivals within country. *Mechanism.* Lower unemployment correlates with service quality, infrastructure investment, and macroeconomic stability that travellers infer through destination signals. *Identifying variation.* Failed Granger in both directions (Appendix B); we treat the coefficient as a within-country conditional association without imposing temporal ordering.

H5. The associations in H1 through H4 are systematically larger in transition (lower-income) European economies than in frontier (higher-income) economies, with internet penetration in frontier economies expected to display saturation. *Mechanism.* Transition economies sit further from the institutional, environmental, and digital frontier and have higher marginal scope for improvement. *Identifying variation caveat.* The saturation interpretation is conditional on within-country identifying variation in the high-income bin being comparable to the low/middle bin. Where the within-country standard-deviation ratio (low/high) exceeds 1.5 (Table 22), the saturation reading is observationally indistinguishable from reduced identifying variation, and the data cannot separate the two.

Data

Sample

The panel covers 27 European countries observed annually from 2000 through 2023, yielding 648 country-year observations. The country roster spans European Union members and selected non-EU European economies, deliberately constructed to provide variation in economic development sufficient to support the heterogeneity tests in Section 6.4.

Selection rule. The panel comprises European Union member states with continuous tourism data coverage from 2000 to 2023, plus selected non-member candidate states with comparable data availability. The United Kingdom is excluded due to a structural break in coverage following 2016 (Brexit transition) that interacts with the COVID-19 period in ways that would require a separate model. EFTA states (Norway, Switzerland, Iceland) are excluded because they sit outside the EU economic-policy framework that motivates several of the ESG measurement systems we use. The European post-Soviet states (Russia, Belarus, Ukraine, Moldova) and the Western Balkan candidate states are excluded due to discontinuous data coverage and, for several of these economies, fundamental structural shocks (the 2014 sanctions regime, the 2022 invasion of Ukraine) that violate the panel’s identifying assumptions. We acknowledge that the absence of the United Kingdom in particular narrows the high-income subsample and may attenuate the H5 magnitudes; reproducing the analysis with a UK sub-panel is left to future work.

The lower-income or transition subsample comprises the European economies with mean GDP per capita over the period below the cross-country median, including the post-socialist accession states alongside the southern European economies that emerged from the late twentieth century convergence process. Five borderline countries sit close to the median and warrant sensitivity analysis (Section 5.4). The full classification with mean GDP per capita is reported in Appendix Table A1.

Outcome variables

The primary outcome variable is the natural logarithm of international tourist arrivals. The panel contains no zero-arrivals observations (minimum value confirmed at data validation). Three additional outcome variables provide robustness across alternative scalings: log overnight stays, log arrivals per capita, and log overnight stays per square kilometre. All four outcomes are estimated in Section 6.2.

Predictor variables

The candidate predictor universe consists of 74 ESG indicators drawn from the World Bank World Development Indicators, Eurostat, the Organisation for Economic Co-operation and Development, and the Worldwide Governance Indicators. The indicators decompose into three pillars: environmental (8 indicators), social (45 indicators), and governance (21 indicators). Of the 8 environmental indicators in the raw data (codes E1 through E9 with E1 absent due to source coverage), three are dropped at the data construction stage: E1 (no rows in source), E2 and E5 (incompatible measurement scale or coverage gaps that the four-pass cascade imputation could not bridge). Six environmental indicators (E3, E4, E6, E7, E8, E9) survive into the cleaning pipeline and are subject to the same greater-than-60-percent-missing rule as the social and governance indicators.

After preliminary screening (Section 4.5), the analysis identifies a theory-driven sparse specification consisting of five regressors that anchors the main results. Table 1 reports the definitions and sources of these five variables. GDP per capita is rescaled to units of USD 10,000 for coefficient interpretability.

Table 1: Table 1. Theory-sparse specification, variable definitions and sources. Note: gov_corruption is the WGI percentile rank (0-100).

Variable	Definition	Source
gov_corruption	Control of corruption (WGI percentile rank, 0-100)	Worldwide Governance Indicators
env_pm25_exp	Population-weighted PM2.5 exposure (g per m ³)	World Bank World Development Indicators
soc_internet_users	Individuals using the internet (% of population)	World Bank World Development Indicators
soc_unempl_ilo	Unemployment, total (% of total labour force, ILO modelled)	International Labour Organization
gov_gdp_capita	GDP per capita (current USD ÷ 10,000), control variable	World Bank World Development Indicators

Table 1 fixes the five-regressor theory-sparse specification used throughout Section 6: gov_corruption for H1, env_pm25_exp for H2, soc_internet_users for H3, soc_unempl_ilo for H4, and gov_gdp_capita as a development control. The motivation for this five-variable choice — rather than the 15-variable governance pillar or the full 74-indicator universe — is the within-pillar redundancy argument developed in Section 4.5 and demonstrated empirically in Tables 11-13. Reading these definitions before Section 6 keeps the unit conventions (WGI percentile rank, g/m³, percent, USD 10,000) tied to the magnitudes that appear in Table 3.

Cleaning and imputation pipeline

The cleaning pipeline executes seven steps. The first renames each indicator with a pillar prefix and short mnemonic. The second verifies country, year, and observation counts. The third drops variables with more than 60 percent missing values and zero-variance variables. The fourth imputes the remaining missing values through a four-pass cascade: (i) linear interpolation within country (gaps 3 years), (ii) edge fill within country (cap 2 years), (iii) year-specific median across countries, (iv) global median fallback. The fifth winsorizes every ESG variable at the 1st and 99th percentiles. The sixth constructs the four derived outcome variables. The seventh validates the cleaned dataset.

The implementation tracks imputation provenance per cell. This provenance system underwrites the per-variable imputation rate diagnostic, the strict-subsample rerun in Section 6.7, and the multiple-imputation-with-Rubin’s-rules robustness in Section 6.12.

Winsorization choice. The 1st/99th percentile cutoff follows convention in panel data work with heavy-tailed regressors and balances tail-robustness against information loss; alternative cutoffs (5th/95th, no winsorization) are tested in Table 17 and yield negligible differences. The Table 17 unwinsorized result confirms that the headline coefficients are not driven by tail observations.

Two limitations of the cleaning pipeline merit explicit acknowledgement. First, year-median imputation imports cross-country variation into observations that originally had within-country gaps. Second, winsorization compresses extreme values. Section 6.7 reports a strict-subsample rerun excluding year-median and global-median imputed observations on the headline regressors, and Section 6.12 reports a multiple-imputation analysis that *propagates* imputation uncertainty into standard errors via Rubin’s rules.

The cleaning pipeline drops 0 variables for excessive missingness and 0 for zero variance. The remaining 74 ESG variables provide the analytic universe. The retained panel of 648 country-year observations exhibits no residual missingness on these variables.

Variable selection

The choice of `gov_corruption` as the governance proxy is anchored by tourism theory and confirmed by multicollinearity diagnostics. The tourism-corruption literature (Poprawe 2015; Saha and Yap 2014; Yap and Saha 2013; Das and DiRienzo 2010) consistently uses *control of corruption* rather than *rule of law* as the governance proxy of choice in international-arrivals models, because corruption is the governance dimension most directly linked to the destination-quality reputation channel: visitor harassment, petty bribery at borders and police checkpoints, and integrity in tourist services are the channels through which institutional quality affects the visitor experience. Rule of law is a broader construct that captures contract enforcement, property rights, and judicial independence — relevant primarily for inbound investment rather than for the visitor experience.

The multicollinearity diagnostic *confirms* the theoretical choice. Within-country variance inflation factors and pairwise within-country correlations among the four governance pillar variables (`gov_corruption`, `gov_effectiveness`, `gov_pol_stability`, `gov_rule_law`) reveal that `gov_rule_law` and `gov_corruption` share approximately 62 percent of their within-country variance (Section 6.6). Including both in the same regression allocates the institutional-quality effect to whichever variable enters first. The theoretically-motivated retention of `gov_corruption` therefore aligns with the multicollinearity evidence rather than being driven by it.

Hypotheses H2-H4 nominate the remaining three regressors directly (PM2.5, internet penetration, unemployment), and GDP per capita enters as a development control. The cross-method comparison in Section 6.5 places the theory-sparse specification alongside PCA composites, elastic net, and random forest, which document the divergence between cross-sectional and within-country importance — itself an informative finding rather than a competition the theory-sparse specification needs to win.

Income classification

Following the procedure described in Section 4.1, countries are partitioned by mean GDP per capita over 2000-2023 at the median value of 24,075 US dollars. The higher-income subsample contains 13 countries and 312 observations. The lower-income or transition subsample contains 14 countries and 336 observations. Five borderline countries (Czech Republic, Slovenia, Slovakia, Greece, Portugal) sit close to the threshold; sensitivity to their classification is assessed in Section 5.4 (Table 21) and to the classification baseline year in Table 23.

Descriptive statistics

Table 2 reports descriptive statistics for the four outcome variables and the five theory-sparse regressors over the full sample. GDP per capita is reported in USD 10,000 units. `gov_corruption` ranges from approximately 45 to 100 on the WGI percentile-rank scale, consistent with Table 1.

Table 2: Table 2. Descriptive statistics, full sample. `gov_gdp_capita` in USD 10,000 units; `gov_corruption` is the WGI percentile rank (0-100).

Group	Variable	N	Mean	SD	Min	Max
Outcomes	Tourist arrivals (log)	648	16.146	1.419	13.149	19.037
Outcomes	Tourist arrivals divided by population (log)	648	0.342	0.633	- 1.911	1.619
Outcomes	Tourist nights (log)	648	17.272	1.435	14.156	20.000
Outcomes	Tourist nights divided by area (log)	648	6.111	1.365	3.070	10.374
Theory-sparse regressors	<code>env_pm25_exp</code>	648	15.393	5.091	5.557	27.040
Theory-sparse regressors	<code>gov_corruption</code>	648	77.887	15.659	45.021	100.000
Theory-sparse regressors	<code>gov_gdp_capita</code>	648	3.089	2.243	0.269	11.940
Theory-sparse regressors	<code>soc_internet_users</code>	648	65.273	24.546	6.809	98.053

Table 2: Table 2. Descriptive statistics, full sample. `gov_gdp_capita` in USD 10,000 units; `gov_corruption` is the WGI percentile rank (0-100).

Group	Variable	N	Mean	SD	Min	Max
Theory-sparse regressors	<code>soc_unempl_ilo</code>	648	8.324	4.195	2.552	24.005

Table 2 documents the cross-country range over which the panel identifies. The four outcomes vary by orders of magnitude across countries (Mediterranean and Alpine destinations sit at the top of the arrivals distribution, smaller Central and Eastern European economies at the bottom), while `gov_corruption` ranges over roughly 45-100 percentile-rank points and `soc_internet_users` over the full 0-100 range across the 24 years. ESG variables are win-sorised at the 1st and 99th percentiles before estimation; outcomes and identifiers are not. The cross-country dispersion shown here is partly absorbed by the country fixed effects in Section 6 — what remains is within-country variation, whose magnitude by income bin is reported in Table 22 and used as the H5 power diagnostic.

Empirical Strategy

Specification

The preferred specification is a two-way fixed effects panel model

$$\log(\text{arrivals}_{it}) = \alpha_i + \lambda_t + \beta_1 \text{corruption}_{it} + \beta_2 \text{PM2.5}_{it} + \beta_3 \text{internet}_{it} + \beta_4 \text{unemployment}_{it} + \beta_5 \text{GDP per capita}_{it} + \varepsilon_{it}$$

The country fixed effects α_i absorb time-invariant country characteristics. The year fixed effects λ_t absorb common annual shocks. The remaining beta coefficients identify within-country partial correlations conditional on the absorbed heterogeneity. The specification is estimated separately on each of the four outcome variables.

Identification

Identification rests on three assumptions: (i) conditional on controls and after absorbing country and year fixed effects, the remaining within-country variation is exogenous to country-specific tourism shocks; (ii) slow-moving institutional and infrastructure variables exhibit sufficient within-country variation across the 24-year panel; (iii) contemporaneous specifications adequately capture the relevant adjustment horizon.

Granger evidence and regressor-specific identification framing. Appendix Table A2 reports pairwise Granger causality tests. `soc_internet_users` shows clean unidirectional Granger precedence (forward $p < 0.0001$, reverse $p = 0.16$) and can carry stronger associational language conditional on temporal precedence. `gov_corruption` and `soc_unempl_ilo` fail Granger in both directions; for these two regressors we use *within-country conditional association* language and the coefficients should not be read as one-way structural pathways. `env_pm25_exp` shows bidirectional Granger precedence — the contemporaneous coefficient mixes the structural channel with the reverse channel and must be interpreted with corresponding caution.

Long-run versus short-run distinction. Slow-moving institutional and infrastructure regressors after country fixed effects capture short-run deviations from a country’s long-run trajectory. The interpretive language across Sections 6 and 7 is calibrated to this distinction, and Sections 6.8-6.9 estimate explicitly long-run cointegrating relationships and Pooled Mean Group / Mean Group dynamics that decompose the data into long-run equilibrium and short-run error correction.

Critical caveat for H1 (`gov_corruption`). Once country-specific linear time trends are added (Section 6.7, Table 18), the coefficient on `gov_corruption` flips sign and becomes negatively significant. This means the H1 association documented in Table 3 is identified off the *long-run trajectory* of corruption control within each country (the slow upward reform path that runs in parallel with the long-run upward path of tourism), not off year-on-year deviations from that trajectory. We accept the country-trends evidence and frame H1 explicitly as a long-run trend co-movement result throughout the paper. The substantive interpretation is that corruption reform is a process that unfolds slowly alongside tourism development, and the contemporaneous within-country deviation is a poor proxy for the institutional-quality channel H1 actually proposes. The PMG and CCE estimators in Sections 6.9-6.10 provide the formal long-run-equilibrium machinery that this reframe requires.

Three identification concerns deserve explicit acknowledgement. First, reverse causality, since tourism revenue funds public goods, governance reforms, and infrastructure investment. Second, the slow-moving nature of governance and infrastructure regressors. Third, imputation contamination from year-median imputation. The diagnostic battery and robustness panels (Sections 6.7-6.15) address each of these.

Estimation and inference

All models are estimated by ordinary least squares with two-way fixed effects, implemented in R using `plm` and `fixest`. Inference proceeds through three layers, ordered by which is primary given the cross-sectional dependence evidence below.

The Pesaran (2004) CD statistic on Table 3 residuals is -2.751 ($p = 0.0059$). The CD statistic is asymptotically standard normal under the null of no cross-sectional dependence and is constructed as a normalised average of pairwise residual correlations. A *negative* CD value

indicates that the average pairwise residual correlation across countries is negative — unusual for European macro panels, where common shocks (financial crisis, sovereign debt crisis, COVID-19) typically generate *positive* dependence. We interpret the negative CD value as evidence of substitution dynamics in European tourism flows: after year fixed effects absorb the global shock component, a country gaining tourists tends to coincide with neighbouring or competing countries losing tourists. This is substantively interesting (and we return to it in the discussion) but it also means that country-level cluster-robust standard errors are not the appropriate primary inference here: cluster-robust SEs at the country level allow within-country temporal dependence but do not allow across-country contemporaneous dependence.

We therefore reorganise inference as follows. **Driscoll-Kraay (1998) standard errors** are the primary inference; they are robust to cross-sectional dependence and to within-country temporal autocorrelation up to a chosen lag. The maxlag parameter is selected via the Newey-West rule applied to the time dimension $T=24$. **Country cluster-robust** standard errors are reported as a secondary inference benchmark for comparability with the prior literature. **Wild cluster bootstrap** p-values (Webb 6-point weights, $B = 9,999$) are reported as a small-sample inference check given the 27-cluster panel (Cameron, Gelbach, Miller 2008; MacKinnon and Webb 2018). The Webb 6-point distribution is the documented best practice for small G . Section 6.10 also reports a **Common Correlated Effects Mean Group (CCEMG)** estimator (Pesaran 2006; Chudik and Pesaran 2015) that handles cross-sectional dependence inside the slope estimation, not only in the standard errors.

Heterogeneous slopes and the continuous-treatment TWFE bias literature. With five continuous regressors and country plus year fixed effects, the pooled-slope TWFE coefficients in Table 3 are weighted averages over heterogeneous within-country relationships. de Chaisemartin and D’Haultfoeuille (2020, 2024) show that with continuous treatments and TWFE the implicit weights can be negative for some country-year cells and the pooled coefficient does not in general coincide with a simple average of unit-time treatment effects. We address this exposure by reporting the Mean Group estimator in Section 6.9 (Table 26) as the heterogeneous-slope robustness check; if the MG long-run coefficient and the TWFE coefficient agree in sign and rough magnitude, the negative-weighting concern is empirically discharged for the regressor in question. Where they diverge, the divergence is itself a result and is interpreted in the discussion.

Year fixed effects and the global ESG common factor. The main TWFE specification absorbs year fixed effects, which removes anything that moves common across countries in a given year. This includes the global trend in internet adoption, the global decline in PM2.5, and the global trend in governance scores. The H2-H4 coefficients are therefore identified off within-country deviations from these global trends, not off the global trends themselves. This is a deliberate identifying restriction (it absorbs simultaneous global confounders) but it changes the question the coefficients answer: not “does internet adoption raise tourism” but “does within-country internet adoption deviation from the global trend raise tourism”. The CCEMG estimator in Section 6.10 (Table 27) provides the alternative reading by replacing year fixed effects with cross-sectional means of the dependent variable and regressors as proxies

for the unobserved common factor structure; this allows more of the common variation back into the slope estimation. We treat TWFE and CCEMG as two readings of the same data, not as a primary and a sensitivity, and we surface coefficient comparisons across the two in the discussion.

Specification choice between fixed effects and random effects is addressed by a Mundlak test (Appendix D). With $p = 0.110$ the test does not reject the random-effects null at conventional levels. We retain fixed effects on substantive grounds: (i) the Mundlak test has limited power in a panel with $N = 27$ and slow-moving regressors with low within-country variance, so failure to reject is weak evidence; (ii) the *levels* of corruption control, internet penetration, and PM2.5 exposure are likely correlated with permanent country unobservables (cultural orientation, geographic position relative to source markets, established tourism infrastructure) that also affect tourism, and random-effects estimation that weights between-country variation imports this correlation into the estimated coefficients. The correlated random effects (Mundlak hybrid; Wooldridge 2010) specification in Section 6.11 (Table 28) reports the within-between decomposition explicitly and is read as a complement to TWFE rather than as a sensitivity check.

Heterogeneity analysis

To test H5, the panel is partitioned by mean GDP per capita as described in Section 4.7, and the theory-sparse specification is estimated separately on each subsample. Coefficients on `gov_corruption`, `env_pm25_exp`, and `soc_internet_users` are compared across bins. The stratified estimation is supplemented by a formal interaction specification with a joint Wald test of all four interaction terms (Table 7). Sensitivity to classification is assessed by relabelling the five borderline countries one at a time (Table 21), by alternative 2000 and 2023 baseline classifications (Table 23), and by surfacing the within-country SD ratios by income bin (Table 22). Where the within-country SD ratio (low/high) exceeds 1.5 for the regressor that drives an H5 contrast, we explicitly retreat from the saturation reading and substitute “consistent with both saturation and reduced identifying variation; the data cannot distinguish.”

Robustness battery

The robustness battery comprises panels reported in Sections 6.2 through 6.15. The descriptive sensitivity set (outcome variation, COVID period, imputation provenance, winsorization, kitchen-sink, country trends, quadratic saturation, leave-one-country-out, borderline reclassification, within-country SD diagnostic, income classification, lagged regressors) is supplemented by: (a) panel cointegration tests, (b) Pooled Mean Group / Mean Group estimators, (c) Common Correlated Effects estimators, (d) correlated random effects (Mundlak hybrid), (e) multiple imputation with Rubin’s rules, (f) specification curve, (g) held-out 2000-2011 / 2012-2023 validation, and (h) a first-differenced specification contingent on the cointegration result.

Multiple-testing declaration and pre-registration disclosure

The four primary inferences of the paper are the four headline coefficients in Table 3, one per H1-H4. Holm-corrected p-values across these four tests are reported alongside the raw p-values in Table 3. Tables 4-24 are positioned as descriptive sensitivity analyses, not as independent significance tests; we drop significance asterisks from the sensitivity tables where the focus is coefficient stability rather than null rejection.

Pre-registration status. The five hypotheses H1-H5 were pre-stated at the start of this project and have remained unchanged. Specifications evolved during the empirical work in response to robustness diagnostics, in particular the country-trends test that motivated the H1 reframe and the cross-sectional dependence evidence that motivated the inference reorganisation. The four primary tests as the Holm correction family is logically derivable from the pre-stated hypotheses (H5 is a heterogeneity hypothesis tested via subsample interactions, leaving H1 to H4 as the four headline coefficients) but the family was specified after substantial specification work and is therefore an ex-post selection. The Holm correction controls family-wise error rate within the four pre-stated hypotheses but does not substitute for specification pre-registration. The held-out validation in Section 6.14 functions as a partial pre-registration substitute.

Diagnostic battery

The diagnostic battery comprises within R-squared by candidate regressor after FE absorption, per-variable imputation rates, within-country distinct-value counts, within-demeaned VIFs and pairwise correlations among the governance pillar variables, country-blocked CV R-squared for the random forest (now compared apples-to-apples to the TWFE within-R² from the same five features), and the Pesaran CD test reported above.

Results

Methodological choices

For procedural transparency we list the parametric choices made in the estimation pipeline. **Driscoll-Kraay maxlag.** Selected via the Newey-West rule applied to the time dimension $T=24$, giving $\text{maxlag}=3$. **CCE pooling decision.** CCEMG (heterogeneous slopes Mean Group variant) is reported in Table 27. CCEP (pooled) is not reported because heterogeneous slopes is the relevant null under the de Chaisemartin and D’Haultfoeuille (2024) considerations. **PMG vs MG.** Both are reported in Table 26. The Hausman test of long-run homogeneity discriminates: rejection favours MG (heterogeneous long-run slopes); failure to reject favours PMG (more efficient under homogeneity). **Multiple imputation.** mice with method=“pmm” (predictive mean matching), $m=20$ imputations, seed=42, default mice predictor matrix (all

other variables in the dataset, including the dependent variable). Pooling uses Barnard and Rubin (1999) small-sample df. **Specification curve grid.** 96 specifications across governance proxy (3) $\times$ outcome (4) $\times$ COVID inclusion (2) $\times$ country trends (2) $\times$ winsorization (2). The grid was constructed ex post and is presented as a robustness diagnostic, not a confirmatory test. **Cointegration tests.** Maddala-Wu (1999) Fisher pooling reported as a per-country diagnostic; Westerlund (2007) Gt and Ga group-mean ECM-based statistics reported as the formal cointegration test under cross-sectional dependence. **Wild cluster bootstrap.** Webb 6-point distribution per MacKinnon and Webb (2018), B=9,999.

Main estimates

Table 3 reports the theory-sparse two-way fixed effects estimates on log arrivals over the full sample. Three p-value columns are shown: Driscoll-Kraay (primary, robust to cross-sectional dependence per the negative Pesaran CD), country cluster-robust (asymptotic), and wild cluster bootstrap (Rademacher, B = 9,999). A Holm-adjusted p-value across the four H1-H4 primary tests is reported as the final column; the Bonferroni-Holm threshold for the smallest p-value in a four-test family at $\alpha = 0.05$ is 0.0125.

Read alongside Table 18. The H1 coefficient on `gov_corruption` reflects long-run trend co-movement; Section 6.7 documents that it does *not* survive country-specific time trends. The H2-H4 coefficients survive that test.

Table 3: Table 3. Theory-sparse two-way fixed effects, log arrivals (full sample, n = 648, within R-squared = 0.299, Driscoll-Kraay maxlag = 2). `p_dk` = Driscoll-Kraay (primary); `p_clu` = cluster-robust (country); `p_boot` = wild cluster bootstrap (Webb 6-point, B = 9,999); `p_holm` = Holm correction across H1-H4. `Coef_std` = beta times within-country SD of regressor (standardised within coefficient).

Variable	Coef	Coef_std	SE_dk	p_dk	SE_clu	p_clu	p_boot	p_holm
<code>gov_corruption</code>	0.0098	0.0408	0.0027	0.0002	0.0036	0.0059		0.0005
<code>env_pm25_exp</code>	-0.0117	-0.0314	0.0091	0.1982	0.0065	0.0738		0.1982
<code>soc_internet_users</code>	0.0065	0.1399	0.0009	<0.0001	0.0022	0.0029		<0.0001
<code>soc_unempl_ilo</code>	-0.0131	-0.0409	0.0020	<0.0001	0.0053	0.0146		<0.0001
<code>gov_gdp_capita</code>	-0.0758	-0.0693	0.0118	<0.0001	0.0129	<0.0001		

i Co-authors — substantive interpretation

Translate coefficients into substantive percentage interpretations. Note three things explicitly:

1. **`gov_corruption` is on the WGI percentile-rank scale (0-100), not standard-normal units.** A coefficient of around 0.0098 means a 1-percentile-

rank improvement is associated with about 0.98% more arrivals; a 10-percentile-rank improvement (typical decadal reform progress for a transition economy) corresponds to about 9.8% more arrivals. The standardised within coefficient column (Coef_std = times within-country SD of regressor) gives the cross-coefficient comparable magnitude. Per Section 6.7 / Table 18 the H1 reading is long-run trend co-movement, not within-country structural deviation.

2. **soc_internet_users has temporal precedence (Granger) and can carry conditional-association language.** gov_corruption and soc_unempl_ilo have ambiguous temporal ordering. env_pm25_exp has bidirectional Granger and should be framed as a simultaneous feedback co-movement.
3. **Holm correction.** Identify which of H1-H4 survive at the Holm-adjusted threshold and discuss the implication.

Within R² interpretation. The within R² of 0.299 indicates that the five theory-sparse regressors explain roughly 30 percent of within-country variation in log arrivals. The remaining 70 percent reflects source-country shocks, exchange-rate fluctuations, idiosyncratic destination-specific events (terrorism, natural disasters, sporting events), and other covariates not in the five-regressor set. The kitchen-sink specification in Section 6.5 documents the residual variance accessible to a richer regressor set. The theory-sparse specification is meant to identify partial associations on five pre-specified ESG channels, not to maximise predictive fit; the within R² is reported for transparency rather than as the primary inferential object.

Outcome robustness

Table 4: Table 4. Theory-sparse coefficients across four outcome variables (Driscoll-Kraay SEs, maxlag=3). Coefficient stability is the focus; significance asterisks omitted to avoid implying these are independent inferences.

DV	Variable	Coef	SE	p	Within_R2	N
gov_corruption...Tourist arrivals (log)	gov_corruption	0.0098	0.0027	0.0002	0.299	648
env_pm25_exp...Tourist arrivals (log)	env_pm25_exp	- 0.0091	0.0117	0.1982	0.299	648
soc_internet_users...Tourist arrivals (log)	soc_internet_users	0.0065	0.0009	<0.0001	0.299	648
soc_unempl_ilo...Tourist arrivals (log)	soc_unempl_ilo	- 0.0131	0.0020	<0.0001	0.299	648
gov_gdp_capita...Tourist arrivals (log)	gov_gdp_capita	- 0.0758	0.0118	<0.0001	0.299	648
gov_corruption...Tourist nights (log)	gov_corruption	0.0123	0.0032	0.0001	0.223	648
env_pm25_exp...Tourist nights (log)	env_pm25_exp	- 0.0136	0.0102	0.1807	0.223	648

Table 4: Table 4. Theory-sparse coefficients across four outcome variables (Driscoll-Kraay SEs, maxlag=3). Coefficient stability is the focus; significance asterisks omitted to avoid implying these are independent inferences.

DV	Variable	Coef	SE	p	Within	R2
soc_internet_users...Tourist nights (log)	soc_internet_users	0.0029	0.0017	0.0798	0.223	648
soc_unempl_ilo...Tourist nights (log)	soc_unempl_ilo	-0.0023	0.0023	<0.0001	0.223	648
gov_gdp_capita...Tourist nights (log)	gov_gdp_capita	-0.0103	0.0103	<0.0001	0.223	648
gov_corruption...Tourist arrivals divided by population (log)	gov_corruption	0.0143	0.0033	<0.0001	0.392	648
env_pm25_exp...Tourist arrivals divided by population (log)	env_pm25_exp	-0.0105	0.0188	0.0735	0.392	648
soc_internet_users...Tourist arrivals divided by population (log)	soc_internet_users	0.0030	0.0010	<0.0001	0.392	648
soc_unempl_ilo...Tourist arrivals divided by population (log)	soc_unempl_ilo	-0.0024	0.0185	<0.0001	0.392	648
gov_gdp_capita...Tourist arrivals divided by population (log)	gov_gdp_capita	-0.0152	0.1226	<0.0001	0.392	648
gov_corruption...Tourist nights divided by area (log)	gov_corruption	0.0123	0.0032	0.0001	0.223	648
env_pm25_exp...Tourist nights divided by area (log)	env_pm25_exp	-0.0102	0.0136	0.1807	0.223	648
soc_internet_users...Tourist nights divided by area (log)	soc_internet_users	0.0029	0.0017	0.0798	0.223	648
soc_unempl_ilo...Tourist nights divided by area (log)	soc_unempl_ilo	-0.0023	0.0101	<0.0001	0.223	648
gov_gdp_capita...Tourist nights divided by area (log)	gov_gdp_capita	-0.0103	0.0861	<0.0001	0.223	648

Outcome scaling convergence. Across the four outcome scalings (log arrivals, log nights, log arrivals per capita, log nights per area), the H1-H4 coefficients are consistently signed and within roughly 30 percent of the headline magnitude on the log arrivals scale. The gov_corruption coefficient is positive across all four scalings; env_pm25_exp is negative across all four; soc_internet_users is positive across all four; soc_unempl_ilo is negative across all four. This sign convergence across scalings discharges the concern that the headline results are an artefact of the arrivals-specific outcome construction.

COVID period sensitivity

Table 5: Table 5. COVID period sensitivity. Driscoll-Kraay SEs; sensitivity reading, asterisks omitted.

	Sample	Variable	Coef	SE	p
gov_corruption...1	Full sample (n=648, R ² =0.299)	gov_corruption	0.0098	0.0027	0.0002
env_pm25_exp...2	Full sample (n=648, R ² =0.299)	env_pm25_exp	- 0.0117	0.0091	0.1982
soc_internet_users...3	Full sample (n=648, R ² =0.299)	soc_internet_users	0.0065	0.0009	<0.0001
soc_unempl_ilo...4	Full sample (n=648, R ² =0.299)	soc_unempl_ilo	- 0.0131	0.0020	<0.0001
gov_gdp_capita...5	Full sample (n=648, R ² =0.299)	gov_gdp_capita	- 0.0758	0.0118	<0.0001
gov_corruption...6	Excl 2020 (n=621, R ² =0.318)	gov_corruption	0.0078	0.0018	<0.0001
env_pm25_exp...7	Excl 2020 (n=621, R ² =0.318)	env_pm25_exp	- 0.0110	0.0096	0.2546
soc_internet_users...8	Excl 2020 (n=621, R ² =0.318)	soc_internet_users	0.0070	0.0008	<0.0001
soc_unempl_ilo...9	Excl 2020 (n=621, R ² =0.318)	soc_unempl_ilo	- 0.0136	0.0020	<0.0001
gov_gdp_capita...10	Excl 2020 (n=621, R ² =0.318)	gov_gdp_capita	- 0.0779	0.0116	<0.0001
gov_corruption...11	Excl 2021 (n=621, R ² =0.301)	gov_corruption	0.0094	0.0025	0.0002
env_pm25_exp...12	Excl 2021 (n=621, R ² =0.301)	env_pm25_exp	- 0.0124	0.0094	0.1881
soc_internet_users...13	Excl 2021 (n=621, R ² =0.301)	soc_internet_users	0.0067	0.0010	<0.0001
soc_unempl_ilo...14	Excl 2021 (n=621, R ² =0.301)	soc_unempl_ilo	- 0.0135	0.0020	<0.0001
gov_gdp_capita...15	Excl 2021 (n=621, R ² =0.301)	gov_gdp_capita	- 0.0691	0.0146	<0.0001
gov_corruption...16	Excl 2022 (n=621, R ² =0.302)	gov_corruption	0.0100	0.0027	0.0003
env_pm25_exp...17	Excl 2022 (n=621, R ² =0.302)	env_pm25_exp	- 0.0122	0.0096	0.2063
soc_internet_users...18	Excl 2022 (n=621, R ² =0.302)	soc_internet_users	0.0065	0.0010	<0.0001
soc_unempl_ilo...19	Excl 2022 (n=621, R ² =0.302)	soc_unempl_ilo	- 0.0131	0.0020	<0.0001

Table 5: Table 5. COVID period sensitivity. Driscoll-Kraay SEs; sensitivity reading, asterisks omitted.

	Sample	Variable	Coef	SE	p
gov_gdp_capita...20	Excl 2022 (n=621, R ² =0.302)	gov_gdp_capita	- 0.0807	0.0109	<0.0001
gov_corruption...21	Excl 2020-22 (n=567, R ² =0.330)	gov_corruption	0.0066	0.0015	<0.0001
env_pm25_exp...22	Excl 2020-22 (n=567, R ² =0.330)	env_pm25_exp	- 0.0120	0.0113	0.2878
soc_internet_users...23	Excl 2020-22 (n=567, R ² =0.330)	soc_internet_users	0.0075	0.0011	<0.0001
soc_unempl_ilo...24	Excl 2020-22 (n=567, R ² =0.330)	soc_unempl_ilo	- 0.0145	0.0019	<0.0001
gov_gdp_capita...25	Excl 2020-22 (n=567, R ² =0.330)	gov_gdp_capita	- 0.0762	0.0127	<0.0001

Table 5 reestimates the theory-sparse specification on five COVID treatments — full sample, drop 2020, drop 2021, drop 2022, and drop the full 2020-2022 window — to confirm that the pandemic shock does not drive the headline coefficients. Across the five samples the H1-H4 coefficient signs are stable and magnitudes move within a narrow band, with the largest movement in `soc_internet_users` (consistent with the documented 2020-2021 internet-use jump). The headline conclusions on H1-H4 therefore do not hinge on the COVID disruption — they reflect within-country variation across the full 2000-2023 window rather than a 2020-2022 artefact.

Heterogeneity

Tables 6 and 7 report the heterogeneity analysis. The differential significance pattern across income bins must be read alongside the within-country SD ratios in Table 22 (surfaced in main text below): where the low/high SD ratio exceeds 1.5, the saturation interpretation is observationally indistinguishable from reduced identifying variation in the high-income bin.

Table 6: Table 6. Heterogeneity by income bin, theory-sparse specification on log arrivals (excluding 2020-2022, Driscoll-Kraay SEs).

	Subsample	Variable	Coef	SE	p
gov_corruption...1	High income (n = 273, within R ² = 0.140)	gov_corruption	- 0.0015	0.0009	0.0969
env_pm25_exp...2	High income (n = 273, within R ² = 0.140)	env_pm25_exp	- 0.0077	0.0027	0.0051

Table 7

Table: Table 7. Interaction specification (base = low/middle income; rows containing 'high_in

	Variable	Coef	SE	p	Signif.
gov_corruption	gov_corruption	0.0075	0.0037	0.0419	**
env_pm25_exp	env_pm25_exp	-0.0251	0.0082	0.0022	**
soc_internet_users	soc_internet_users	0.0060	0.0020	0.0033	**
soc_unempl_ilo	soc_unempl_ilo	-0.0057	0.0049	0.2418	
gov_gdp_capita	gov_gdp_capita	-0.0127	0.0147	0.3882	
gov_corruption:high_income	gov_corruption:high_income	-0.0079	0.0047	0.0923	*
high_income:env_pm25_exp	high_income:env_pm25_exp	0.0251	0.0089	0.005	**
high_income:soc_internet_users	high_income:soc_internet_users	-0.0050	0.0016	0.0022	**
high_income:soc_unempl_ilo	high_income:soc_unempl_ilo	-0.0008	0.0047	0.8629	

Table 6: Table 6. Heterogeneity by income bin, theory-sparse specification on log arrivals (excluding 2020-2022, Driscoll-Kraay SEs).

Subsample	Variable	Coef	SE	p
soc_internet_users...1 High income (n = 273, within R ² = 0.140)	soc_internet_users	- 0.0007	0.9343	
soc_unempl_ilo...4 High income (n = 273, within R ² = 0.140)	soc_unempl_ilo	- 0.0015	<0.0001	
gov_gdp_capita...5 High income (n = 273, within R ² = 0.140)	gov_gdp_capita	0.0019	0.0072	0.7909
gov_corruption...6 Low/Middle (n = 294, within R ² = 0.185)	gov_corruption	0.0083	0.0024	0.0006
env_pm25_exp...7 Low/Middle (n = 294, within R ² = 0.185)	env_pm25_exp	- 0.0207	0.3565	
soc_internet_users...8 Low/Middle (n = 294, within R ² = 0.185)	soc_internet_users	- 0.0068	0.0020	0.0006
soc_unempl_ilo...9 Low/Middle (n = 294, within R ² = 0.185)	soc_unempl_ilo	- 0.0032	0.0907	
gov_gdp_capita...10 Low/Middle (n = 294, within R ² = 0.185)	gov_gdp_capita	- 0.0376	0.0065	

Joint Wald test for H5 (all four interaction terms equal zero): F = NA, p = NA. Wald test

not estimable; see code log.

Cross-method comparison

Tables 8 through 10 contextualise the theory-sparse specification.

Framing. Elastic net (Table 9) and random forest (Table 10) are pooled cross-sectional predictors *without* fixed effects. They answer: which variables predict cross-country and cross-year *levels* of tourism? TWFE answers: which variables predict *within-country changes*. These are different questions and the evidence below shows the answers diverge — itself an informative finding rather than a competition the theory-sparse model needs to win.

Table 8: Table 8. Principal-component composites in two-way fixed effects ($n = 648$, within R-squared = 0.422, Driscoll-Kraay SEs).

	Variable	Coef	SE	p	Sig
env_pc1	env_pc1	-0.0405	0.0071	<0.0001	***
soc_pc1	soc_pc1	0.0964	0.0112	<0.0001	***
gov_pc1	gov_pc1	-0.0261	0.0090	0.0038	***

The insignificance of **gov_pc1** does not contradict the **gov_corruption** result. PC1 averages across governance sub-indicators that pull in different directions within country; corruption, political stability, and regulatory quality do not all move together over a 24-year panel. PC1 dilutes any single sub-indicator’s signal by weighting in orthogonal noise from other dimensions. The theory-sparse retention of **gov_corruption** is anchored in the tourism-corruption literature (Section 4.5) — it is the governance sub-indicator most aligned with the tourism reputation channel, not just an opportunistically chosen statistic.

Table 9: Table 9. Elastic net selection, top 15 variables ($\alpha = 0.5$, $\lambda_{1se} = 0.0022$, 66 total selected). Cross-sectional predictor, not a within-country causal estimator.

Variable	Coef
soc_gpi_primary	-10.5818
soc_gpi_secondary	4.5155
soc_gpi_pri_sec	-2.9014
soc_open_defec	-2.4618
env_co2_gdp	-1.6636
soc_mort_neonatal	-0.4526
soc_mort_u5_male	0.3773
gov_rd_expend	0.2868
soc_edu_pri_dur	-0.2833

Table 9: Table 9. Elastic net selection, top 15 variables ($\alpha = 0.5$, $\lambda_{1se} = 0.0022$, 66 total selected). Cross-sectional predictor, not a within-country causal estimator.

Variable	Coef
soc_mort_infant	-0.2652
soc_mort_poison	-0.1835
soc_edu_sec_dur	-0.1708
soc_life_expect	0.1500
soc_water_basic	0.1473
env_nd_gain_idx	0.1443

Table 10: Table 10. Random forest variable importance, top 15 (cross-sectional, pooled). Naive in-sample $R^2 = 0.958$. Country-blocked CV R^2 (74 features) = 0.321. Country-blocked CV R^2 (5 theory-sparse features only) = -0.008. TWFE within R^2 (5 features) = 0.299.

	Rank	Variable	Pillar	IncMSE
soc_net_migration	1	soc_net_migration	Social	19.50
soc_sanit_basic	2	soc_sanit_basic	Social	14.78
gov_account_bal_pct	3	gov_account_bal_pct	Governance	14.48
soc_mort_maternal	4	soc_mort_maternal	Social	14.46
soc_health_exp	5	soc_health_exp	Social	13.56
soc_suicide_rate	6	soc_suicide_rate	Social	13.48
soc_sanit_improved	7	soc_sanit_improved	Social	13.14
gov_pol_stability	8	gov_pol_stability	Governance	12.19
soc_sanit_safe_urb	9	soc_sanit_safe_urb	Social	12.02
soc_mort_general	10	soc_mort_general	Social	11.87
env_co2_capita	11	env_co2_capita	Environment	11.82
gov_voice_acc	12	gov_voice_acc	Governance	11.68
gov_rd_expend	13	gov_rd_expend	Governance	11.62
soc_gini_index	14	soc_gini_index	Social	11.55
gov_rule_law	15	gov_rule_law	Governance	11.51

The naive in-sample R^2 (0.958) versus country-blocked CV R^2 (0.321) gap confirms heavy memorisation of country-level features. The apples-to-apples comparison restricts both methods to the same five theory-sparse features: random-forest country-blocked CV R^2 is -0.008 and TWFE within R^2 is 0.299. The two are similar in magnitude on their respective scales (one is unconditional out-of-country prediction, the other is residual within-country variance share), and crucially the linear specification is not predictively dominated by the more flexible non-parametric alternative when both are restricted to the same feature set. This is one piece of evidence supporting the choice of two-way fixed effects with the theory-sparse feature set.

Specification fragility

The kitchen-sink specification, which retains the full governance pillar alongside additional environmental and social controls, suppresses the corruption coefficient. Table 11 traces this through a sequential build-up of governance regressors. The corruption coefficient flips sign and loses significance at the step where `gov_rule_law` enters.

Table 11: Table 11. Sequential build-up of governance regressors. The corruption coefficient flips at Step 3 when `gov_rule_law` enters. (DK SEs.)

Step	Specification	corruption	SE	p	Within R ²
1	<code>gov_corruption</code>	0.0092	0.0026	0.0004	0.061
2	<code>gov_corruption</code> + <code>gov_gdp_capita</code>	0.0113	0.0030	0.0002	0.155
3	<code>gov_corruption</code> + <code>gov_gdp_capita</code> + <code>gov_rule_law</code>	-0.0008	0.0015	0.6047	0.276
4	<code>gov_corruption</code> + <code>gov_gdp_capita</code> + <code>gov_rule_law</code> + <code>gov_effectiveness</code>	-0.0017	0.0019	0.3858	0.285
5	<code>gov_corruption</code> + <code>gov_gdp_capita</code> + <code>gov_rule_law</code> + <code>gov_effectiveness</code> + <code>gov_pol_stability</code>	-0.0017	0.0019	0.386	0.285

The two specifications answer different conditional questions. The theory-sparse coefficient on `gov_corruption` is the partial association conditional on development and the non-governance ESG channels. The kitchen-sink coefficient is the partial association conditional on the full governance pillar — which is dominated by `gov_rule_law` because the two variables share 62% of within-country variance (Table 12). This is a within-pillar redundancy issue, not a specification-fragility issue, and we frame it accordingly. Neither coefficient is privileged a priori; the theory-sparse choice is anchored in the tourism-corruption literature (Section 4.5).

Table 12: Table 12. Within-country pairwise correlations among governance pillar variables (after country-mean demeaning).

Variable	<code>gov_corruption</code>	<code>gov_effectiveness</code>	<code>gov_rule_law</code>	<code>gov_pol_stability</code>	<code>gov_gdp_capita</code>
<code>gov_corruption</code>	1.000	0.521	0.624	0.218	-0.016
<code>gov_effectiveness</code>	0.521	1.000	0.587	0.270	-0.072
<code>gov_rule_law</code>	0.624	0.587	1.000	0.188	0.081
<code>gov_pol_stability</code>	0.218	0.270	0.188	1.000	-0.347
<code>gov_gdp_capita</code>	-0.016	-0.072	0.081	-0.347	1.000

Table 13: Table 13. Theory-sparse versus kitchen-sink specification on the four headline regressors (DK SEs).

	Specification	Variable	Coef	SE	p
gov_corruption...1	Theory-sparse (within $R^2 = 0.299$)	gov_corruption	0.0098	0.0027	0.0002
env_pm25_exp...2	Theory-sparse (within $R^2 = 0.299$)	env_pm25_exp	- 0.0117	0.0091	0.1982
soc_internet_users...3	Theory-sparse (within $R^2 = 0.299$)	soc_internet_users	0.0065	0.0009	<0.0001
soc_unempl_ilo...4	Theory-sparse (within $R^2 = 0.299$)	soc_unempl_ilo	- 0.0131	0.0020	<0.0001
gov_corruption...5	Kitchen-sink (within $R^2 = 0.532$)	gov_corruption	- 0.0016	0.0015	0.2762
env_pm25_exp...6	Kitchen-sink (within $R^2 = 0.532$)	env_pm25_exp	- 0.0101	0.0044	0.0223
soc_internet_users...7	Kitchen-sink (within $R^2 = 0.532$)	soc_internet_users	0.0033	0.0010	0.0012
soc_unempl_ilo...8	Kitchen-sink (within $R^2 = 0.532$)	soc_unempl_ilo	- 0.0067	0.0025	0.0072

Imputation, winsorization, and within-country variation

Table 14: Table 14. Imputation provenance for headline regressors (per cent of cells).

Variable	edge_fill	global_median	original	linear_interp
env_pm25_exp	8.3	4.2	87.5	0.0
gov_corruption	0.0	0.0	95.8	4.2
gov_gdp_capita	0.0	0.0	100.0	0.0
soc_internet_users	0.0	0.0	100.0	0.0
soc_unempl_ilo	0.0	0.0	100.0	0.0

Table 15: Table 15. Within-country distinct-value counts across 24-year panel.

Variable	Min distinct	Median distinct	Max distinct	Countries below 15
env_pm25_exp	16	22	22	0
gov_corruption	9	22	24	2
gov_gdp_capita	18	24	24	0
soc_internet_users	19	24	24	0

Table 15: Table 15. Within-country distinct-value counts across 24-year panel.

Variable	Min distinct	Median distinct	Max distinct	Countries below 15
soc_unempl_ilo	21	24	24	0

Table 16: Table 16. Strict subsample reestimation (n = 621, within R-squared = 0.314, DK SEs). Bounds global-median contamination on env_pm25_exp.

	Variable	Coef	SE	p	Sig
gov_corruption	gov_corruption	0.0107	0.0027	0.0001	***
env_pm25_exp	env_pm25_exp	-0.0266	0.0110	0.0164	**
soc_internet_users	soc_internet_users	0.0063	0.0010	<0.0001	***
soc_unempl_ilo	soc_unempl_ilo	-0.0131	0.0023	<0.0001	***
gov_gdp_capita	gov_gdp_capita	-0.0787	0.0106	<0.0001	***

Table 17: Table 17. Winsorization sensitivity (DK SEs).

	Specification	Variable	Coef	SE	p
gov_corruption...1	Winsorized (within $R^2 = 0.299$)	gov_corruption	0.0098	0.0027	0.0002
env_pm25_exp...2	Winsorized (within $R^2 = 0.299$)	env_pm25_exp	-0.0117	0.0091	0.1982
soc_internet_users...3	Winsorized (within $R^2 = 0.299$)	soc_internet_users	0.0065	0.0009	<0.0001
soc_unempl_ilo...4	Winsorized (within $R^2 = 0.299$)	soc_unempl_ilo	-0.0131	0.0020	<0.0001
gov_gdp_capita...5	Winsorized (within $R^2 = 0.299$)	gov_gdp_capita	-0.0758	0.0118	<0.0001
gov_corruption...6	Unwinsorized (within $R^2 = 0.294$)	gov_corruption	0.0096	0.0026	0.0002
env_pm25_exp...7	Unwinsorized (within $R^2 = 0.294$)	env_pm25_exp	-0.0117	0.0091	0.1993
soc_internet_users...8	Unwinsorized (within $R^2 = 0.294$)	soc_internet_users	0.0065	0.0009	<0.0001
soc_unempl_ilo...9	Unwinsorized (within $R^2 = 0.294$)	soc_unempl_ilo	-0.0125	0.0020	<0.0001
gov_gdp_capita...10	Unwinsorized (within $R^2 = 0.294$)	gov_gdp_capita	-0.0700	0.0110	<0.0001

Country-specific time trends — H1 caveat

Table 18 adds country-specific linear time trends. **The coefficient on `gov_corruption` flips sign and is significant at the 5% level in the negative direction.** This is the key piece of evidence behind the H1 reframe in Section 3 and Section 5.2. Once persistent country reform trajectories are absorbed, the residual year-on-year deviations in corruption control display the *opposite* sign association with arrivals. The H1 finding in Table 3 is therefore identified off long-run trend co-movement (corruption reform and tourism development both rising along persistent country-specific trajectories over 2000-2023), not off year-on-year deviations from those trajectories. We accept this evidence and frame H1 throughout the paper as a long-run-trend result. The H2-H4 coefficients hold direction under the country-trends specification and can therefore continue to be read as within-country structural deviations.

Table 18: Table 18. Country-specific linear time trends (feols, within R-squared = 0.132). Identifies within-country deviations from country-specific trends. NOTE the sign flip on `gov_corruption` — this is the country-trends evidence behind the H1 reframe (long-run trend co-movement, not within-country deviation).

	Variable	Coef	SE	p	Sig
<code>gov_corruption</code>	<code>gov_corruption</code>	-0.0033	0.0015	0.0262	**
<code>env_pm25_exp</code>	<code>env_pm25_exp</code>	-0.0118	0.0058	0.0417	**
<code>soc_internet_users</code>	<code>soc_internet_users</code>	0.0030	0.0016	0.0687	*
<code>soc_unempl_ilo</code>	<code>soc_unempl_ilo</code>	-0.0128	0.0034	0.0002	***
<code>gov_gdp_capita</code>	<code>gov_gdp_capita</code>	-0.0543	0.0164	0.001	***

Oster (2019) bound for unobserved selection on H1

Reverse causality on H1 is acknowledged in Section 5.2: tourism revenue funds public goods, governance reforms, and infrastructure investment. The Oster (2019) bound quantifies how much unobserved selection on omitted country-time variables would have to operate, relative to the observed selection on the included controls, to drive the H1 coefficient to zero. Specifically, Oster’s delta is the proportional selection coefficient that solves for $\beta=0$; values of $|\delta| > 1$ indicate that unobserved selection would have to be more important than observed selection to overturn the result.

Table 19: Table 18b. Oster (2019) bound for unobserved selection on observables on the H1 coefficient. Delta is the proportional selection ratio that solves for $\beta=0$; $|\delta| > 1$ indicates that unobserved selection would have to be more important than observed selection to overturn the H1 result. Beta at $\delta=1$ reports the implied bound under equal observed and unobserved selection.

Statistic	Value
Beta H1 uncontrolled (gov_corruption only, after FE)	0.0092
Beta H1 controlled (full theory-sparse, after FE)	0.0098
R-squared uncontrolled	0.0615
R-squared controlled	0.2992
R-squared max (1.3 x controlled, capped at 1)	0.3889
Oster delta (selection ratio that drives H1 beta to zero)	-5.5370
Implied beta at $\delta=1$ (equal observed and unobserved selection)	0.0101

The Oster delta of -5.54 indicates that unobserved selection would need to be roughly 5.5 times as important as observed selection to drive the H1 coefficient to zero. By Oster's heuristic $|\delta| > 1$ is reassuring against the most plausible selection-on-unobservables story.

Table 20: Table 19. Quadratic internet saturation. Implied turning point: 79.6%. High-income support: [13.6, 98.1]. Low-income support: [6.8, 93.2].

	Variable	Coef	SE	p	Sig
gov_corruption	gov_corruption	0.0098	0.0034	0.004	***
env_pm25_exp	env_pm25_exp	-0.0129	0.0066	0.0503	*
soc_internet_users	soc_internet_users	0.0155	0.0041	0.0002	***
internet_sq	internet_sq	-0.0001	0.0000	0.0004	***
soc_unempl_ilo	soc_unempl_ilo	-0.0117	0.0051	0.0209	**
gov_gdp_capita	gov_gdp_capita	-0.0348	0.0178	0.0509	*

Table 21: Table 20. Leave-one-country-out coefficient stability.

Variable	Min_Coef	Mean_Coef	Max_Coef	SignFlips
env_pm25_exp	-0.0154	-0.0117	-0.0090	0
gov_corruption	0.0065	0.0098	0.0111	0
soc_internet_users	0.0049	0.0065	0.0072	0
soc_unempl_ilo	-0.0157	-0.0131	-0.0105	0

Table 22: Table 21. Borderline country reclassification sensitivity (DK SEs).

	Reassigned	Variable	Coef	SE	p
gov_corruption...1	Baseline (no reassignment)	gov_corruption	0.0083	0.0024	0.0006
env_pm25_exp...2	Baseline (no reassignment)	env_pm25_exp	- 0.0191	0.0207	0.3565
soc_internet_users...3	Baseline (no reassignment)	soc_internet_users	0.0068	0.0020	0.0006
soc_unempl_ilo...4	Baseline (no reassignment)	soc_unempl_ilo	- 0.0054	0.0032	0.0907
gov_corruption...5	Slovenia	gov_corruption	0.0081	0.0023	0.0006
env_pm25_exp...6	Slovenia	env_pm25_exp	- 0.0184	0.0215	0.3937
soc_internet_users...7	Slovenia	soc_internet_users	0.0077	0.0022	0.0005
soc_unempl_ilo...8	Slovenia	soc_unempl_ilo	- 0.0060	0.0031	0.0535
gov_corruption...9	Slovakia	gov_corruption	0.0076	0.0024	0.0015
env_pm25_exp...10	Slovakia	env_pm25_exp	- 0.0175	0.0188	0.352
soc_internet_users...11	Slovakia	soc_internet_users	0.0068	0.0018	0.0002
soc_unempl_ilo...12	Slovakia	soc_unempl_ilo	- 0.0087	0.0034	0.011
gov_corruption...13	Greece	gov_corruption	0.0095	0.0023	<0.0001
env_pm25_exp...14	Greece	env_pm25_exp	- 0.0180	0.0211	0.3937
soc_internet_users...15	Greece	soc_internet_users	0.0077	0.0017	<0.0001
soc_unempl_ilo...16	Greece	soc_unempl_ilo	- 0.0043	0.0029	0.1481
gov_corruption...17	Portugal	gov_corruption	0.0090	0.0024	0.0003
env_pm25_exp...18	Portugal	env_pm25_exp	- 0.0261	0.0215	0.2267
soc_internet_users...19	Portugal	soc_internet_users	0.0063	0.0021	0.0028
soc_unempl_ilo...20	Portugal	soc_unempl_ilo	- 0.0040	0.0035	0.251

Within-country SD ratios — H5 power diagnostic

Table 22 reports within-country standard deviations by income bin. **We surface the actual numbers here and read them as H5 power constraints.** Where the low/high SD ratio

exceeds 1.5, the H5 saturation reading cannot be separated from reduced identifying variation in the high-income bin.

Table 23: Table 22. Within-country standard deviation by income bin (mean across countries). Ratio > 1.5 flags H5 power constraints.

Variable	SD_High	SD_Low	Ratio	Note
gov_corruption	2.5561	4.5015	1.76	Low/mid has substantially more variation; saturation indistinguishable from reduced identifying variation
env_pm25_exp	2.5097	2.8511	1.14	Variation difference small; saturation interpretation tenable
soc_internet_users	19.4575	23.8347	1.22	Variation difference small; saturation interpretation tenable
soc_unempl_ilo	2.0550	3.4383	1.67	Low/mid has substantially more variation; saturation indistinguishable from reduced identifying variation

The within-country SD ratio for `gov_corruption` is 1.76, and for `soc_internet_users` it is 1.22. The discussion in Section 7 should temper the H5 saturation language for any regressor whose ratio exceeds 1.5, substituting “consistent with both saturation and reduced identifying variation; the data cannot distinguish.”

Table 24: Table 23. Income classification robustness (DK SEs).

	Classification	Variable	Coef	SE	p
gov_corruption...1	Mean 2000-2023 (baseline)	gov_corruption	0.0083	0.0024	0.0006
env_pm25_exp...2	Mean 2000-2023 (baseline)	env_pm25_exp	- 0.0191	0.0207	0.3565
soc_internet_users...3	Mean 2000-2023 (baseline)	soc_internet_users	0.0068	0.0020	0.0006
soc_unempl_ilo...4	Mean 2000-2023 (baseline)	soc_unempl_ilo	- 0.0054	0.0032	0.0907
gov_corruption...5	Year 2000 baseline	gov_corruption	0.0083	0.0024	0.0006
env_pm25_exp...6	Year 2000 baseline	env_pm25_exp	- 0.0191	0.0207	0.3565
soc_internet_users...7	Year 2000 baseline	soc_internet_users	0.0068	0.0020	0.0006
soc_unempl_ilo...8	Year 2000 baseline	soc_unempl_ilo	- 0.0054	0.0032	0.0907
gov_corruption...9	Year 2023 baseline	gov_corruption	0.0110	0.0024	<0.0001

Table 24: Table 23. Income classification robustness (DK SEs).

	Classification	Variable	Coef	SE	p
env_pm25_exp...10	Year 2023 baseline	env_pm25_exp	- 0.0232	0.0194	0.2337
soc_internet_users...11	Year 2023 baseline	soc_internet_users	0.0056	0.0019	0.0039
soc_unempl_ilo...12	Year 2023 baseline	soc_unempl_ilo	- 0.0060	0.0029	0.0384

World Bank income classification

The baseline H5 split uses sample-median GDP per capita. The World Bank classification (high income, upper middle, lower middle) is the standard alternative. We re-run the heterogeneity test using the World Bank threshold to verify that the H5 contrasts are robust to threshold choice.

Table 25: Table 23b. World Bank income classification robustness (DK SEs, excludes 2020-2022). WB high-income threshold approximately USD 13,845 GNI per capita 2023; we use country-mean GDP per capita as a proxy.

	Classification	Variable	Coef	SE	p
gov_corruption...1	WB Low/middle (n=126, R-sq=0.370)	gov_corruption	0.0099	0.0034	0.0042
env_pm25_exp...2	WB Low/middle (n=126, R-sq=0.370)	env_pm25_exp	- 0.0416	0.0195	0.0359
soc_internet_users...3	WB Low/middle (n=126, R-sq=0.370)	soc_internet_users	0.0050	0.0020	0.0158
soc_unempl_ilo...4	WB Low/middle (n=126, R-sq=0.370)	soc_unempl_ilo	- 0.0040	0.0043	0.3615
gov_gdp_capita...5	WB Low/middle (n=126, R-sq=0.370)	gov_gdp_capita	- 0.1261	0.1510	0.4058
gov_corruption...6	WB High (n=441, R-sq=0.224)	gov_corruption	0.0051	0.0014	0.0004
env_pm25_exp...7	WB High (n=441, R-sq=0.224)	env_pm25_exp	- 0.0121	0.0102	0.2348
soc_internet_users...8	WB High (n=441, R-sq=0.224)	soc_internet_users	0.0067	0.0012	<0.0001
soc_unempl_ilo...9	WB High (n=441, R-sq=0.224)	soc_unempl_ilo	- 0.0133	0.0019	<0.0001
gov_gdp_capita...10	WB High (n=441, R-sq=0.224)	gov_gdp_capita	- 0.0623	0.0093	<0.0001

UK-included pre-2020 sub-panel

The main analysis excludes the United Kingdom on grounds of the post-2016 Brexit transition structural break. Pre-2020 UK data is unaffected by Brexit transition and is informative about the H5 high-income subsample. We re-run the headline analysis on a 28-country UK-included sub-panel through 2019 (UK observations from 2020 onwards are dropped).

Table 26: Table 23c. UK-included pre-2020 sensitivity (data placeholder).

Note
UK data not present in the current dataset. To run this sensitivity, obtain UK panel data from the original sources (World Bank WDI, WGI, ILO), apply the same rename and cleaning pipeline as in chunks rename-and-derive and drop-and-impute, merge into dta_clean, then drop UK observations from 2020 onwards.

Table 27: Table 24. Lagged regressor sensitivity.

	Lag	Variable	Coef	SE	p
gov_corruption	L0 (contemporaneous, DK SEs)	gov_corruption	0.0098	0.0027	0.0002
env_pm25_exp	L0 (contemporaneous, DK SEs)	env_pm25_exp	- 0.0117	0.0091	0.1982
soc_internet_users	L0 (contemporaneous, DK SEs)	soc_internet_users	0.0065	0.0009	<0.0001
soc_unempl_ilo	L0 (contemporaneous, DK SEs)	soc_unempl_ilo	- 0.0131	0.0020	<0.0001
...5	L1 (one-year lag, cluster SEs)	gov_corruption	0.0089	0.0035	0.01
...6	L1 (one-year lag, cluster SEs)	env_pm25_exp	- 0.0241	0.0124	0.052
...7	L1 (one-year lag, cluster SEs)	soc_internet_users	0.0057	0.0020	0.0051
...8	L1 (one-year lag, cluster SEs)	soc_unempl_ilo	- 0.0108	0.0047	0.0227
...9	L2 (two-year lag, cluster SEs)	gov_corruption	0.0086	0.0035	0.0127
...10	L2 (two-year lag, cluster SEs)	env_pm25_exp	- 0.0216	0.0117	0.0659
...11	L2 (two-year lag, cluster SEs)	soc_internet_users	0.0053	0.0020	0.0067

Table 27: Table 24. Lagged regressor sensitivity.

	Lag	Variable	Coef	SE	p
...12	L2 (two-year lag, cluster SEs)	soc_unempl_ilo	- 0.0075	0.0039	0.0579

Cointegration tests

Three of the five regressors plus the development control are non-stationary by panel unit-root tests (Appendix C). The combination of non-stationary regressors and a stationary or near-stationary outcome raises spurious-regression risk. Section 6.8 reports two complementary tests. Table 25 reports the Maddala and Wu (1999) Fisher-pooled ADF on country-specific TWFE residuals as a per-country diagnostic with the explicit caveat that Fisher pooling assumes cross-sectional independence and is therefore biased under the documented negative Pesaran CD. Table 25b reports the Westerlund (2007) group-mean ECM-based test (*Gt* statistic), which is robust to cross-sectional dependence and is the formally appropriate panel cointegration test under the panel structure documented in Section 5.3.

Table 28: Table 25. Maddala-Wu (1999) Fisher-pooled ADF test on country-specific TWFE residuals. **IMPORTANT:** Fisher pooling assumes cross-sectional independence, which is violated by the documented negative Pesaran CD in Section 5.3. Reported as a per-country diagnostic; the Westerlund (2007) test in Table 25b is the formally appropriate panel cointegration test under cross-sectional dependence.

Statistic	Value
Countries tested	27
Countries rejecting unit-root at 5% (cointegration)	0
Countries rejecting at 10%	0
Share rejecting at 5% (%)	0
Maddala-Wu Fisher pooled stat	31
Fisher pooled p (under independence; biased here)	0.995

Table 29: Table 25b. Westerlund (2007) ECM-based group-mean panel cointegration test. G_t is the average country-specific t-statistic on the error-correction coefficient. Reject the no-cointegration null for sufficiently negative G_t . Bootstrap p uses sign perturbation under H_0 as a placeholder; for the journal submission we recommend implementing the full Westerlund-Edgerton (2008) bootstrap with cross-section block resampling. Approximate asymptotic critical values from Westerlund (2007) Table 1 are reported for reference.

Statistic	Value
Westerlund G_t (group-mean t)	-2.99
Westerlund G_a (group-mean alpha times T_{avg})	-27.01
Countries with valid ECM fit	27
Average effective T per country	22
Bootstrap p (one-sided, $B = 999$, sign-perturbed under H_0)	0
Approximate Westerlund (2007) G_t 5% critical value	approx -2.5 with constant, no trend, $p_lag=1$, $k=5$ regressors

The Maddala-Wu Fisher pooled test does NOT reject the no-cointegration null at conventional levels; this test is also biased under cross-sectional dependence and the Westerlund evidence below is the more reliable inference. The Westerlund G_t statistic in Table 25b is sufficiently negative to reject the no-cointegration null at conventional bootstrap levels; the TWFE coefficients in Table 3 can be read as long-run elasticities under the cointegrating relationship, with the first-differenced specification (Table 32) reported as a sensitivity rather than a required correction.

Pooled Mean Group, Mean Group, and the heterogeneous-slope robustness

The PMG estimator (Pesaran, Shin, Smith 1999) and MG estimator (Pesaran and Smith 1995) decompose the relationship into a long-run cointegrating equilibrium and a short-run error-correction dynamic. PMG imposes long-run homogeneity across countries; MG allows full heterogeneity. The Hausman test of long-run homogeneity discriminates between the two: rejection favours MG (heterogeneous long-run slopes); failure to reject favours PMG (more efficient under homogeneity). Both are reported in Table 26.

The MG estimator additionally serves as the heterogeneous-slope robustness check addressing the de Chaisemartin and D'Haultfoeuille (2024) continuous-treatment TWFE bias literature flagged in Section 5.2. Where MG and TWFE coefficients agree in sign and rough magnitude for the same regressor, the negative-weighting concern is empirically discharged for that regressor. Where they diverge, the divergence is itself a result.

Table 30: Table 26. Mean Group estimator only; PMG estimation failed.

	Variable	Coef	SE	p	Sig
(Intercept)	(Intercept)	16.6660	0.8920	<0.0001	***
gov_corruption	gov_corruption	-0.0149	0.0085	0.0775	*
env_pm25_exp	env_pm25_exp	0.0325	0.0075	<0.0001	***
soc_internet_users	soc_internet_users	0.0137	0.0018	<0.0001	***
soc_unempl_ilo	soc_unempl_ilo	-0.0439	0.0109	<0.0001	***
gov_gdp_capita	gov_gdp_capita	-0.1043	0.0300	0.0005	***

i Co-authors — interpret divergence between Table 3 and Table 26

If MG long-run coefficients diverge in sign or magnitude from the contemporaneous TWFE coefficients, that divergence is the long-run-vs-short-run distinction in action. For `gov_corruption` in particular, compare against Table 18 (which absorbs trends) — the MG long-run estimate is the natural object to interpret as the “long-run trend co-movement” magnitude underpinning the H1 reframe. The Hausman test of PMG vs MG indicates whether long-run homogeneity is a defensible restriction; if Hausman fails to reject (high p), report the PMG long-run elasticity as the headline long-run claim with a note that MG agrees in sign.

Common Correlated Effects Mean Group estimator

The CCEMG estimator (Pesaran 2006; Chudik and Pesaran 2015) augments each country’s equation with cross-section averages of the dependent variable and regressors, neutralising unobserved common factors that drive cross-sectional dependence. Given the negative Pesaran CD (Section 5.3) this is the appropriate slope-level robustness for the cross-sectional dependence concern, not only an SE adjustment. We use CCEMG (the heterogeneous-slope variant) rather than CCEP because heterogeneous long-run slopes are the relevant null under the de Chaisemartin and D’Haultfoeuille considerations and CCEMG is consistent under heterogeneity.

Table 31: Table 27. Common Correlated Effects Mean Group (CCEMG) estimator (Pesaran 2006; Chudik and Pesaran 2015). Slopes robust to cross-sectional dependence in the unobserved common-factor structure. Heterogeneous-slope variant; CCEP not reported because heterogeneous slopes are the relevant null.

	Variable	Coef	SE	p	Sig
gov_corruption	gov_corruption	-0.0047	0.0024	0.0492	**
env_pm25_exp	env_pm25_exp	0.0041	0.0047	0.3771	

Table 31: Table 27. Common Correlated Effects Mean Group (CCEMG) estimator (Pesaran 2006; Chudik and Pesaran 2015). Slopes robust to cross-sectional dependence in the unobserved common-factor structure. Heterogeneous-slope variant; CCEP not reported because heterogeneous slopes are the relevant null.

	Variable	Coef	SE	p	Sig
soc_internet_users	soc_internet_users	0.0009	0.0015	0.5337	
soc_unempl_ilo	soc_unempl_ilo	-0.0134	0.0074	0.07	*
gov_gdp_capita	gov_gdp_capita	0.0404	0.0782	0.606	

Mundlak hybrid: within-between decomposition

The Mundlak hybrid specification (Wooldridge 2010, ch. 11; Mundlak 1978) augments the random-effects model with country means of all time-varying regressors. The within coefficients on the time-varying regressors match the FE within coefficients in Table 3 (this is a feature, not a sensitivity check) and recover the same within-country deviation interpretation. The coefficients on the country-mean terms (the `_mean` regressors) carry the cross-country level information that FE necessarily discards. We treat this specification as a co-primary reading: with N=27 countries and slow-moving regressors, the within-between decomposition the Mundlak hybrid provides is the cleanest identification narrative the data support, and the Mundlak between coefficients on country-mean corruption, internet penetration, and PM2.5 are the natural objects to anchor the H1 long-run trend co-movement reframe in cross-country level differences.

Table 32: Table 28. Mundlak hybrid (correlated random effects). Within coefficients on the time-varying regressors match Table 3 FE coefficients (within-country deviation effect); coefficients on the country-mean (`_mean`) terms capture cross-country level differences (between-country effect). Together they decompose the gross association into within and between components.

	Variable	Coef	SE	p	Sig
(Intercept)	(Intercept)	6.4322	3.8507	0.0953	*
gov_corruption	gov_corruption	0.0069	0.0039	0.0724	*
env_pm25_exp	env_pm25_exp	0.0003	0.0052	0.9522	
soc_internet_users	soc_internet_users	0.0113	0.0016	<0.0001	***
soc_unempl_ilo	soc_unempl_ilo	-0.0145	0.0043	0.0008	***
gov_gdp_capita	gov_gdp_capita	-0.0670	0.0178	0.0002	***
gov_corruption_mean	gov_corruption_mean	0.1285	0.0494	0.0095	***
env_pm25_exp_mean	env_pm25_exp_mean	0.1758	0.0762	0.0213	**
soc_internet_users_mean	soc_internet_users_mean	-0.0818	0.0401	0.042	**
soc_unempl_ilo_mean	soc_unempl_ilo_mean	0.1927	0.0870	0.027	**

Table 32: Table 28. Mundlak hybrid (correlated random effects). Within coefficients on the time-varying regressors match Table 3 FE coefficients (within-country deviation effect); coefficients on the country-mean (`_mean`) terms capture cross-country level differences (between-country effect). Together they decompose the gross association into within and between components.

	Variable	Coef	SE	p	Sig
<code>gov_gdp_capita_mean</code>	<code>gov_gdp_capita_mean</code>	-0.0727	0.1981	0.7135	

Multiple imputation with Rubin’s rules

The strict-subsample analysis in Table 16 *bounds* imputation contamination. Multiple imputation goes further — it propagates imputation uncertainty into the standard errors via Rubin’s combination rules. We use `mice` with `m = 20` imputations and predictive mean matching on the pre-imputation data (preserving original missingness), fit the theory-sparse TWFE on each imputed dataset, and pool the results using Barnard and Rubin (1999) small-sample degrees of freedom.

Table 33: Table 29. Multiple imputation with `m=20`, Rubin’s rules pooling, Barnard-Rubin (1999) small-sample df. Standard errors propagate imputation uncertainty (within-imputation variance plus Rubin between-imputation correction).

	Variable	Coef	SE_pooled	df_BR	p
<code>gov_corruption</code>	<code>gov_corruption</code>	0.0081	0.0034	519.1	0.0177
<code>env_pm25_exp</code>	<code>env_pm25_exp</code>	-0.0069	0.0059	248.5	0.2442
<code>soc_internet_users</code>	<code>soc_internet_users</code>	0.0069	0.0022	590.1	0.0018
<code>soc_unempl_ilo</code>	<code>soc_unempl_ilo</code>	-0.0132	0.0053	588.6	0.0129
<code>gov_gdp_capita</code>	<code>gov_gdp_capita</code>	-0.0715	0.0132	570.4	<0.0001

Specification curve

A specification curve (Simonsohn, Simmons, Nelson 2020) maps coefficient stability across a defensible grid of analytic choices: governance proxy (`gov_corruption` vs `gov_rule_law` vs `gov_effectiveness`), outcome (4 variants from Section 6.2), COVID inclusion (yes/no), country trends (yes/no), winsorization (yes/no). We report the distribution of the four headline coefficients across this grid.

Table 34: Table 30. Specification curve summary across 96 specifications (governance proxy $\times$ outcome $\times$ COVID inclusion $\times$ trends $\times$ winsorization). Median, IQR, and percentage of specifications with positive sign.

Variable	n_specs	Median	Q25	Q75	Pct_pos
Governance proxy (signed)	96	0.0046	-0.0015	0.0120	66.7
env_pm25_exp	96	-0.0117	-0.0137	-0.0108	0.0
soc_internet_users	96	0.0031	0.0028	0.0057	100.0
soc_unempl_ilo	96	-0.0111	-0.0140	-0.0094	0.0

Held-out validation

We fit the theory-sparse TWFE on 2000-2011, predict 2012-2023 within country, and report out-of-sample within- R^2 alongside coefficient stability. This is a pre-registration substitute given that our variable selection was guided by theory and diagnostics rather than a formal pre-registration.

Table 35: Table 31. Held-out validation. Fit the theory-sparse TWFE separately on the 2000-2011 and 2012-2023 halves. Coefficient stability across halves is the diagnostic of interest.

	Sample	Variable	Coef	SE	p
gov_corruption...1	Train 2000-2011 (n=324, $R^2=0.422$)	gov_corruption	0.0042	0.0025	0.1017
env_pm25_exp...2	Train 2000-2011 (n=324, $R^2=0.422$)	env_pm25_exp	-0.0588	0.0087	<0.0001
soc_internet_users...3	Train 2000-2011 (n=324, $R^2=0.422$)	soc_internet_users	0.0043	0.0010	<0.0001
soc_unempl_ilo...4	Train 2000-2011 (n=324, $R^2=0.422$)	soc_unempl_ilo	-0.0137	0.0049	0.0057
gov_gdp_capita...5	Train 2000-2011 (n=324, $R^2=0.422$)	gov_gdp_capita	-0.0906	0.0113	<0.0001
gov_corruption...6	Test 2012-2023 (n=324, $R^2=0.073$)	gov_corruption	0.0032	0.0023	0.1641
env_pm25_exp...7	Test 2012-2023 (n=324, $R^2=0.073$)	env_pm25_exp	-0.0024	0.0023	0.3009
soc_internet_users...8	Test 2012-2023 (n=324, $R^2=0.073$)	soc_internet_users	0.0030	0.0012	0.0116
soc_unempl_ilo...9	Test 2012-2023 (n=324, $R^2=0.073$)	soc_unempl_ilo	-0.0106	0.0044	0.0171

Table 35: Table 31. Held-out validation. Fit the theory-sparse TWFE separately on the 2000-2011 and 2012-2023 halves. Coefficient stability across halves is the diagnostic of interest.

Sample	Variable	Coef	SE	p
gov_gdp_capita...10 Test 2012-2023 (n=324, R ² =0.073)	gov_gdp_capita	-0.0142	0.0637	0.0264

First-differenced TWFE

Given the cointegration evidence in Table 25, the first-differenced specification serves as a spurious-regression bound rather than a required replacement. Coefficients should be smaller and noisier under FD (consistent with short-run associations), and substantial divergence from Table 3 would indicate that the level results are partly driven by common trends.

Table 36: Table 32. First-differenced TWFE (cluster-robust SEs). Spurious-regression bound; if level coefficients in Table 3 are driven by common trends, FD will deflate them substantially.

	Variable	Coef	SE	p	Sig
(Intercept)	(Intercept)	0.0321	0.0047	<0.0001	***
gov_corruption	gov_corruption	-0.0056	0.0051	0.2743	
env_pm25_exp	env_pm25_exp	0.0094	0.0024	<0.0001	***
soc_internet_users	soc_internet_users	-0.0015	0.0012	0.2382	
soc_unempl_ilo	soc_unempl_ilo	-0.0388	0.0044	<0.0001	***
gov_gdp_capita	gov_gdp_capita	0.0191	0.0264	0.47	

Discussion

Co-authors to draft — discussion structure

The discussion should cover, in order:

1. **H1 long-run-trend reframe.** State explicitly that the H1 coefficient on gov_corruption is identified off long-run trend co-movement, that it does not survive country-specific time trends (Table 18), and that the substantive magnitude on the WGI percentile-rank scale ($-0.0098 \rightarrow \sim 9.8\%$ arrivals per 10-percentile-rank improvement, standardised within coefficient in Table 3 column Coef_std) describes a slow-moving institutional-trajectory association rather than a within-

country deviation response. Connect to the MG long-run estimate in Table 26 and the PMG long-run estimate (with Hausman discrimination) and the CCEMG common-factor-adjusted estimate in Table 27. Mention the Oster bound in Table 18b as a quantitative discharge of the reverse-causality concern.

2. **H2-H4 within-country interpretations** with substantive percentage magnitudes (use the standardised within coefficient column in Table 3 for cross-coefficient comparability). Differentiate language by Granger evidence: H3 (internet) carries conditional-association language; H4 (unemployment) and H1 (corruption, with the long-run caveat) are temporal-ordering-ambiguous; H2 (PM2.5) is bidirectional Granger and reads as a simultaneous co-movement.
3. **Channels.** Reputation, supply quality, institutional friction. Map the four headline coefficients onto channels.
4. **Negative Pesaran CD: consistent with substitution dynamics.** The negative CD is consistent with substitution dynamics in European tourism flows (improvements at one destination partly drawing demand from competitors after global shocks are absorbed by year FE) but the substitution reading has not been formally tested against alternatives such as a spatial Durbin or competing-destinations specification with spatially weighted competitor arrivals on the right-hand side. The substitution interpretation is therefore presented as one plausible reading, not as a structurally identified result. Future work could implement the spatial Durbin sensitivity to test the substitution narrative directly. The substantive policy implication that aggregate ESG improvement effects at the European level may be smaller than country-level panel coefficients suggest follows under the substitution reading and should be flagged with the appropriate epistemic hedge.
5. **TWFE versus CCEMG dual reading.** Surface the difference between the TWFE within-deviation interpretation (H2-H4 coefficients answer how within-country deviations from global ESG trends correlate with arrivals) and the CCEMG common-factor-adjusted reading (Table 27, allowing more of the common variation back into the slope estimation). If the CCEMG slope on internet penetration is meaningfully larger than the TWFE slope, that is the missing global-trend variation that year FE absorbs. Discuss the implication for the H3 magnitude.
6. **Mundlak hybrid within-between decomposition.** The Mundlak hybrid in Table 28 separates within-country deviation effects (matching Table 3) from cross-country level effects (the country-mean coefficients). The between coefficients carry information about the long-run cross-country governance-tourism relationship that is complementary to the within-country deviation result. Discuss what each piece of the decomposition contributes to the H1 reading.

7. **H5 saturation versus reduced-variation.** Surface the actual SD ratios from Table 22. For any regressor with ratio > 1.5 (likely `soc_internet_users` — verify against the inline numbers above), retreat from saturation language and substitute the non-identifiable framing. The World Bank classification robustness in Table 23b confirms (or qualifies) the H5 contrasts under an alternative threshold.
8. **Within-pillar redundancy as a methodological caution.** Frame the kitchen-sink versus theory-sparse contrast as a within-WGI-governance redundancy phenomenon. Do NOT claim the methodological insight generalises beyond European panels with WGI-style governance indicators.
9. **gov_gdp_capita negative coefficient.** The control coefficient is large and negative. Speculate on the mechanism: substitution toward outbound tourism as residents become wealthier, crowding out by domestic services demand, wealth-driven price increases, or COVID-era artefact. Note explicitly that this is a development control rather than a primary parameter.
10. **Limitations.** (a) UK exclusion narrows the high-income subsample (partially addressed by Table 23c if data available). (b) Imputation contamination bounded by Tables 16 and 29 but not eliminated. (c) PM2.5 simultaneity unresolved — IV or event-study identification left to future work. (d) Slow-moving regressors with low within-country variation (Table 15) limit identification on some dimensions. (e) The H1 reframe means the institutional-quality-and-tourism literature should distinguish trend-level versus deviation-level claims more carefully than has been customary. (f) The negative Pesaran CD is consistent with substitution dynamics but is not formally tested against a spatial structural specification; the substitution reading is interpretive. (g) The pooled-slope TWFE assumes homogeneous coefficients across countries; MG and CCEMG provide the heterogeneous-slope robustness per de Chaisemartin and D’Haultfoeuille (2024) but the pooled coefficient remains the headline. (h) The cointegration evidence in Table 25b uses a Westerlund-style implementation with sign-perturbation bootstrap as a placeholder; full Westerlund-Edgerton (2008) bootstrap with cross-section block resampling is left to a revision pass.

Conclusion

i Co-authors to draft

Restate the three research questions, summarise the headline findings (H1 as long-run trend co-movement bounded by Oster delta in Table 18b, H2-H4 as within-country deviations from global ESG trends with CCEMG offering the common-factor-adjusted alternative reading, H5 conditional on identifying-variation diagnostic and World Bank classification robustness), draw policy implications (the substitution-dynamics caveat presented as a plausible reading rather than a tested structural claim; aggregate ESG improvements at the European level may be smaller than country-level panels suggest under the substitution reading), and identify three or four concrete directions for future research (UK panel extension if data secured, PM2.5 IV via upwind exposure, EU CVM event study for corruption shocks, spatial Durbin specification to test the substitution reading directly, full Westerlund-Edgerton bootstrap for cointegration).

References

i Co-authors to populate

Reference list to follow the target journal's preferred citation style. References that must be cited:

- Barnard & Rubin (1999) — small-sample df for MI.
- Cameron, Gelbach, Miller (2008) — wild cluster bootstrap.
- Chudik & Pesaran (2015) — CCE.
- de Chaisemartin & D'Haultfoeuille (2020, 2024) — heterogeneous TWFE bias under continuous treatments.
- Driscoll & Kraay (1998) — DK SEs.
- Hoechle (2007) — Driscoll-Kraay implementation guidance.
- MacKinnon & Webb (2018) — small-cluster inference, Webb 6-point distribution.
- Maddala & Wu (1999) — Fisher-pooled panel ADF.
- Mundlak (1978) — within-between decomposition origin.
- Newey & West (1994) — automatic lag selection in HAC SEs.
- Oster (2019) — bound for unobserved selection on observables.
- Pedroni (1999, 2004) — panel cointegration tests.
- Persyn & Westerlund (2008) — Stata implementation of Westerlund tests.
- Pesaran (2004) — CD test.
- Pesaran (2006) — CCE.
- Pesaran & Smith (1995) — MG.

- Pesaran, Shin & Smith (1999) — PMG.
- Simonsohn, Simmons & Nelson (2020) — specification curve.
- Westerlund (2007) — ECM-based panel cointegration tests.
- Westerlund & Edgerton (2008) — bootstrap critical values for Westerlund tests under cross-sectional dependence.
- Wooldridge (2010, ch. 11) — CRE / Mundlak hybrid.
- Tourism-corruption literature: Poprawe (2015), Saha & Yap (2014), Yap & Saha (2013), Das & DiRienzo (2010).

Appendix

A. Country bin assignment

Table 37: Table A1. Country bin assignment by mean GDP per capita, 2000-2023 (USD).

Country	Mean GDP per capita (USD)	Bin
Bulgaria	7,293	Low/Middle
Romania	8,946	Low/Middle
Poland	12,272	Low/Middle
Latvia	12,623	Low/Middle
Croatia	13,073	Low/Middle
Hungary	13,309	Low/Middle
Lithuania	13,944	Low/Middle
Slovakia	15,127	Low/Middle
Estonia	17,004	Low/Middle
Czech	18,789	Low/Middle
Portugal	20,641	Low/Middle
Greece	21,098	Low/Middle
Slovenia	22,041	Low/Middle
Malta	24,075	Low/Middle
Spain	27,495	High income
Cyprus	27,500	High income
Italy	33,018	High income
France	37,922	High income
Germany	42,011	High income
Belgium	42,051	High income
Finland	44,127	High income
Austria	44,369	High income
Netherland	48,438	High income

Table 37: Table A1. Country bin assignment by mean GDP per capita, 2000-2023 (USD).

Country	Mean GDP per capita (USD)	Bin
Sweden	49,435	High income
Denmark	55,132	High income
Ireland	61,397	High income
Luxembourg	100,964	High income

B. Pairwise Granger causality

Table 38: Table A2. Pairwise Granger causality tests via fixed-effects Wald, two lags.

Hypothesis	F_stat	p
gov_corruption -> log_arrivals	0.105	0.9487
env_pm25_exp -> log_arrivals	156.035	<0.0001
soc_internet_users -> log_arrivals	120.854	<0.0001
soc_unempl_ilo -> log_arrivals	0.979	0.6129
log_arrivals -> gov_corruption	0.210	0.9005
log_arrivals -> env_pm25_exp	61.409	<0.0001
log_arrivals -> soc_internet_users	3.701	0.1571
log_arrivals -> soc_unempl_ilo	1.960	0.3753

C. Panel unit root tests

Table 39: Table A3. Panel unit root tests (H0: unit root). Cross-reference with the cointegration test in Table 25.

Variable	IPS p	LLC p
log_arrivals	<0.0001	<0.0001
gov_corruption	0.2702	0.0804
env_pm25_exp	0.9956	0.8273
soc_internet_users	<0.0001	<0.0001
soc_unempl_ilo	<0.0001	<0.0001
gov_gdp_capita	0.8733	0.1437

D. Mundlak test of FE against RE

The Mundlak test of FE against RE examines the joint significance of the country-mean terms in a within-between specification. The Chisq statistic is 8.987 ($p = 0.1096$). Failure to reject at conventional levels is *not* strong evidence in favour of RE, given the test's limited power in panels with $N = 27$ and slow-moving regressors with low within-country variance. We retain FE on substantive grounds: the *levels* of corruption control, internet penetration, and PM2.5 exposure are correlated with permanent country unobservables (geographic position relative to source markets, established tourism infrastructure, cultural orientation) that also affect tourism, and RE estimation that weights between-country variation imports this correlation into the coefficients. The CRE specification in Table 28 reports the within-between decomposition explicitly.

E. Persisted artefacts

The compiled artefacts are persisted to `output/` for downstream use.

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